WNDRVR
SUPPORT NETWORK

VXWORKS 7 SERIAL OVER BLUETOOTH TUTORIAL, SR0650

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VxWorks 7 Serial over Bluetooth Tutorial, SR0650

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1. CREATING A VXWORKS TO HOST SERIAL CONNECTION OVER BLUETOOTH ON A BEAGLEBONE BLACK TARGET

The VxWorks® 7 Serial over Bluetooth Tutorial describes the use of a Bluetooth modem module to create a VxWorks host-to-serial connection over Bluetooth without any software modification.

Before You Begin

Before beginning this tutorial, you must have the following:

- A BeagleBone Black target board.
- A SparkFun Bluetooth Mate Silver.

  Note: This tutorial was tested with the silver model but you may be able to use another model such as Gold or BlueSMiRF.

- Jumper wires, a breadboard, and other support items as needed.
- A host with Bluetooth serial port profile (SPP) support.

Note: The instructions in this tutorial assume you are using an Ubuntu host. However, the tutorial has been tested on Ubuntu and Windows 7. For Ubuntu, you should install the Blueman Bluetooth Manager to simplify the setup. You may also need to uninstall or disable the modem manager as it can prevent you from using **/dev/refcomm0**.

About This Task

Prepare the BeagleBone Black board with a VxWorks image that is configured to autoboot.

Note: The default baud rate for the Bluetooth Mate is 115200, which is the same as the default value for the BSP. If you change the baud rate of the BSP, you must also change the module's setting.

Wire the BeagleBone Black board and the Bluetooth Mate according to the following settings:

Bluetooth Mate	BeagleBone Black
RX-I	Serial debug header pin 5.
TX-O	Serial debug header pin 4.
VCC	Expansion header P9 pin 3.
GND	Expansion header P9 pin 1 (GND).

Note: Use an AC adapter, not an SB cable to power the board. This keeps the Bluetooth module alive even when the target board is powered off.

Procedure

1. Power up the target.

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The red LED on the Bluetooth Mate begins blinking.

2. If it is not already enabled, enable Bluetooth on the host.

   You can use the Blueman Bluetooth Manager for enabling and disabling Bluetooth.

3. Select **Setup New Device** from the Blueman menu.
4. Select the module from the device list.

   The default device name is **RN42-XXXX**.

5. Select **Proceed Without Pairing**.
6. Select **Connect to: "Serial Port"**.
7. When prompted, type **1234** as the passcode. (This is the default.)

   The host is now connected to the Bluetooth module. The file is **/dev/rfcomm0**, **/dev/rfcomm1**, and so forth. The red LED is off and the yellow LED is lit on the module.

8. Open a serial connection using a command (for example, cu) or a GUI (for example, **gtkterm**).

   The baud rate does not matter on the host side for the rfcomm devices. (For example, 'cu -l /dev/rfcomm0 -s 115200'.)

Results

If it is configured in your VxWorks image, you can now use the VxWorks target shell. You can also press the reset button on the BeagleBone Black target to see the boot dialogs and so forth without losing the Bluetooth connection.